

EMBEDDED SYSTEMS

MINOR PROJECT: Digital Thermometer using Arduino

Step-by-step guidance (10 points):

- Define measurement range and accuracy requirements.
- Select temperature sensor such as LM35 or DS18B20.
- Design circuit diagram with Arduino and sensor.
- Assemble hardware components on breadboard.
- Write Arduino code for sensor data acquisition.
- Calibrate sensor readings.
- Display temperature on LCD or serial monitor.
- Test system under different temperature conditions.
- Analyze accuracy and response time.
- Document circuit, code, and results.

Dataset details:

- Dataset name: Temperature Sensor Readings
- Dataset source: Real-time sensor data
- Dataset location: Arduino Serial Monitor CSV
- Dataset link:
 - <https://www.kaggle.com/datasets/sohier/temperature-time-series-dataset>

Tools and platforms:

- Arduino UNO
- LM35 / DS18B20 sensor
- Arduino IDE
- LCD 16x2

Reference links:

- Arduino temperature GitHub: <https://github.com/RobTillaart/Arduino>
 - Sensor interfacing GitHub: https://github.com/adafruit/Adafruit_Sensor
 - Embedded basics GitHub: <https://github.com/arduino/Arduino>
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MAJOR PROJECT: Smart Home Automation System

Step-by-step guidance (10 points):

- Define automation scope such as lighting, fans, and security.
- Select microcontroller platform such as ESP32 or Arduino.
- Choose communication protocol (Wi-Fi/Bluetooth).
- Design system architecture and wiring.
- Interface sensors and actuators.
- Develop firmware for device control.
- Set up cloud or local control server.
- Develop mobile or web control interface.
- Implement automation rules and schedules.
- Test system reliability and scalability.

Dataset details:

- Dataset name: IoT Smart Home Sensor Dataset
- Dataset source: Real-time IoT sensor data
- Dataset location: Cloud database
- Dataset links:
 - <https://www.kaggle.com/datasets/uciml/electricity-load-diagrams-20112014>

Tools and platforms:

- ESP32 / Arduino
- Relay modules
- Firebase / Blynk / MQTT
- Android Studio

Reference links:

- Smart home GitHub: <https://github.com/home-assistant/core>
- IoT automation GitHub: <https://github.com/ESP32-Hub>
- MQTT GitHub: <https://github.com/eclipse/paho.mqtt.python>